Customer Engagement & Product Utilization Analytics for Retention Strategy

Project Report

Abstract

Customer churn is an important challenge for businesses that aim to maintain long-term customer relationships. This project analyses customer engagement, product utilization, account balances, and customer exit using a dataset containing 10,000 customer records. Microsoft Excel 2019 was used to organize the data, create classifications, calculate churn rates, construct PivotTables, and develop a dashboard. The overall churn rate was 20.37%, representing 2,037 customers marked as exited. Single-product customers had a higher observed churn rate than multi-product customers. Churn rates also varied across product-count categories and engagement profiles. These findings can help identify customer segments for further investigation and support data-informed retention planning. The analysis is descriptive and does not establish that any individual factor causes churn.

Keywords: Customer churn, customer engagement, product utilization, customer retention, Excel analytics, dashboard.

1. Introduction

Customer retention is an important aspect of business performance because losing customers can affect revenue, customer relationships, and future growth. Understanding the characteristics of customers who leave can help organizations identify patterns and investigate opportunities to improve customer experience.

Customer churn analysis involves examining customer records to determine how exit rates differ across groups. Product usage, customer activity, and account balances may provide useful information when comparing customer segments.

This project uses Excel-based data analysis to examine customer churn and engagement patterns. The findings are summarized through PivotTables, charts, and a dashboard to make the results easier to understand.

2. Objectives

The main objectives of this project are:

- To calculate churn rates according to the number of products used.
- To compare churn among single-product and multi-product customers.
- To examine average account balances by customer activity status.
- To explore the relationship between a custom salary-balance classification and churn.
- To compare churn across customer engagement profiles.
- To present the findings in a dashboard that can support further retention analysis.

3. Dataset and Tools

The dataset contains 10,000 customer records and includes fields such as customer identifier, credit score, geography, gender, age, tenure, account balance, number of products, estimated salary, activity status, and exit status.

Microsoft Excel 2019 was used for data analysis and visualization.

The main tools and techniques included:

- Excel formulas, including IF and AND.

- Helper columns for customer classification.

- PivotTables for summarizing customer groups.

- Churn-rate calculations.

- Charts and dashboard visualization.

4. Methodology

4.1 Data organization

The customer dataset was organized in Excel, and relevant fields were used to summarize customer activity, product utilization, account balances, and exit status.

4.2 Churn-rate calculation

Churn rate was calculated using the following formula:

Churn Rate = (Number of Exited Customers / Total Customers in the Group) × 100

This calculation was applied to the overall dataset and to the different customer segment

4.3 Product utilization analysis

Customers were grouped according to the number of products used. A separate comparison was made between single-product customers and multi-product customers, with multi-product customers defined as those using two or more products.

4.4 Activity and balance analysis

Average account balances were compared between active and inactive customers using the activity-status field.

4.5 Salary-balance classification

An exploratory classification was created using the dataset's average estimated salary and average balance. Customers with estimated salaries above the dataset average and balances below the dataset average were classified as “Mismatch”; the remaining records were classified as “Normal.”

This classification was created solely for exploratory analysis. It does not indicate fraud, financial misconduct, or the reason a customer exited.

4.6 Engagement-profile analysis

Customers were assigned to engagement profiles using activity status, number of products, and balance relative to the dataset median. Churn rates were then compared across the resulting groups.

4.7 Dashboard development

The results were summarized in a dashboard showing key customer counts, overall churn, product-related churn, average balance by activity status, salary-balance classification, and churn by engagement profile.

5. Results

5.1 Overall customer churn

The dataset contained 10,000 customers, of whom 2,037 were marked as exited.

The overall churn rate was 20.37%. The remaining 7,963 records were marked as not exited.

5.2 Churn by number of products

Number of products	Customers	Exited customers	Churn rate
1	5,084	1,409	27.71%
2	4,590	348	7.58%
3	266	220	82.71%
4	60	60	100.00%

The highest observed churn rates occurred among customers using three or four products. However, these groups contained only 266 and 60 customers, respectively. Their high rates should therefore be interpreted with caution.

5.3 Single-product versus multi-product customers

Customer group	Customers	Exited customers	Churn rate
Single Product	5,084	1,409	27.71%
Multi Product	4,916	628	12.77%

Single-product customers had a higher observed churn rate than multi-product customers. This difference suggests that product utilization may be a useful area for further investigation, but the analysis does not prove that using more products directly reduces churn.

5.4 Average balance by activity status

Activity status	Customers	Average balance
Inactive	4,849	77,134.38
Active	5,151	75,875.42

The average balances of active and inactive customers were relatively close. Activity status alone should not be used to infer the reason for customer exit.

5.5 Salary-balance classification and churn

Classification	Customers	Exited customers	Churn rate
Mismatch	2,011	309	15.37%
Normal	7,989	1,728	21.63%

The Normal group had a higher observed churn rate than the Mismatch group. Because the classification was created using two dataset averages, this result should be treated as exploratory and should not be interpreted as a causal relationship.

5.6 Churn by engagement profile

Engagement profile	Churn rate
Active Engaged	9.06%
Active Low-Product	18.97%
Inactive Disengaged	21.21%
Inactive High-Balance	37.33%

The Active Engaged profile had the lowest observed churn rate, while the Inactive High-Balance profile had the highest. These differences can help guide further investigation into customer needs, activity, and service usage.

6. Discussion

The analysis shows that churn varied across the customer groups examined. Single-product customers had a substantially higher observed churn rate than multi-product customers. The engagement-profile analysis also showed differences between active and inactive customer segments.

These findings suggest that product utilization and customer activity may be useful dimensions for customer-retention analysis. However, the results should be interpreted in the context of the dataset, group sizes, and the definitions used to create each classification.

The high churn rates among customers using three or four products require particular caution because the groups are comparatively small. Similarly, the salary-balance classification is an exploratory grouping rather than an established measure of customer risk.

Further analysis using additional customer characteristics, historical records, and validated business context would be needed to understand the reasons behind these patterns.

7. Recommendations

Based on the findings, the following actions may be considered:

- Investigate single-product customers: Examine their needs, product experience, and reasons for leaving before designing targeted retention measures.
- Review inactive customer segments: Explore whether inactive customers face service, engagement, or product-related barriers.
- Investigate high-churn product groups: Study customers using three or four products in greater detail, taking their small group sizes into account.
- Improve customer engagement: Evaluate appropriate communication and support strategies for customer segments with higher observed churn.
- Monitor churn regularly: Use updated data and dashboards to track changes in customer exits and retention outcomes.
- Validate before taking action: Combine these descriptive findings with additional evidence before making customer-level decisions.

8. Limitations

This project has several limitations.

First, the analysis is based on the available dataset and does not independently establish why customers exited. Second, the results describe associations and group differences rather than causal effects. Third, the three-product and four-product groups are relatively small, which limits how confidently their churn rates can be generalized.

The engagement profiles and salary-balance classifications were created for this project and should not be treated as validated industry standards. The findings may not generalize to other datasets, organizations, or time periods.

9. Conclusion

This project used Microsoft Excel 2019 to analyse customer engagement, product utilization, account balances, and churn across 10,000 customer records. The overall churn rate was 20.37%. Single-product customers had a churn rate of 27.71%, compared with 12.77% among multi-product customers. Churn also differed across engagement profiles, with the highest observed rate in the Inactive High-Balance group.

The dashboard provides a clear summary of these patterns and demonstrates how Excel formulas, PivotTables, and charts can support customer analytics. The findings offer starting points for further investigation and retention planning, while recognizing that additional evidence is required to determine the causes of customer churn.

10. Project Deliverables

The project deliverables include the Excel workbook containing the dataset, analysis formulas, PivotTables, charts, and customer churn dashboard, along with this written project report.

Document status: This document is a project report prepared to describe an Excel-based analytics project. It has not been represented as a peer-reviewed, approved, or published research paper.